

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack features (Next.js + Firebase/GCP) for a provider-facing **SMART on FHIR** clinical decision support app that surfaces covered, guideline-backed medication alternatives inside the EHR prescribing workflow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend that builds alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage, grounded, evidence-first AI rather than an opaque model.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer with full technical ownership delegated by the PI, building the lab's CCR (Contextualized Construct Representations) text-analysis platform end to end (Python, **FastAPI**, sentence-transformers, React) and productionizing a published NLP method into a self-serve web tool for researchers.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend suite to **40 tests** (hermetic CI).

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant eTriage service stable.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research agent (TypeScript) that fetches web sources and streams multi-step research digests over WebSocket, with durable workflows (Durable Objects + WorkflowEntrypoint + D1) that survive process restarts and individual tool failures.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a cross-platform desktop application (React, Electron, FastAPI) with a fully offline on-device pipeline (MediaPipe hand tracking, gesture classification, whisper.cpp ASR, local TTS), designing async APIs and queues to keep the UI responsive across the full stack.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, C++, Java, SQL, Bash, MongoDB, PostgreSQL, SQLite

Frameworks & Web: React, Next.js, Node.js, Express, FastAPI, REST APIs, Docker, Git, GitHub Actions

AI & Agents: LLM agents, RAG, MCP, PyTorch, Hugging Face Transformers, sentence-transformers, evaluation

Systems & Perf: Async services, queues, multi-tenant architecture, Linux profiling, Spark, distributed systems

Cloud & Platforms: AWS, GCP, Azure, Linux, Kubernetes, MongoDB Atlas

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.